ContextForge: Architecture-First Persistent Agentic Memory via Model Context Protocol

Information-Theoretic Security, OR-Set CRDT Sync, Tri-Core Failover, Differential Context Injection, and a 22-Tool MCP Server

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Abstract

Stateless Retrieval-Augmented Generation (RAG) systems suffer from three structural failure modes: zero adversarial block rate, high cold-start failover latency (∼480 ms), and unconstrained context injection that grows linearly with session history. We present **ContextForge**, a five-pillar agentic memory architecture that closes all three gaps simultaneously and exposes every capability as a **22-tool Model Context Protocol (MCP)** server—making persistent, adversarially-hardened memory natively available in Claude Desktop, Cursor, VS Code, and Windsurf without any client-side code changes.

The central contribution is the *Entropy-Gated Soft-Gate Ledger*: an append-only SQLite event store whose `ReviewerGuard` filters writes via dual signals—Shannon entropy H and Lempel–Ziv compression density ρ—catching obfuscated payloads ($H > H^* = 3.5$ bits) and repetition attacks ($\rho < 0.60$) independently. Verified-origin traffic receives an elevated threshold $H^*_{\text{VOH}} \approx 4.375$ bits via HMAC-SHA256 tokens. A Sliding-Window Temporal Correlator extends detection to slow-drip multi-write escalation sequences, achieving 100% detection at 0% false-positive rate. A trigram perplexity gate (Pass 0.5, calibration threshold $P^* = 231.8$) provides a third independent defence against entropy-mimicry payloads, which score $> 2 \times P^*$ under the reference model.

A new OR-Set CRDT synchronisation layer (`src/sync/crdt_sync.py`) enables conflict-free concurrent writes from multiple IDE clients with add-wins semantics, verified at 100% convergence across three concurrent-write scenarios including a full split-brain reconnection.

Across 530 benchmark tests, ContextForge achieves: +90.0 pp adversarial block rate, F1 = 1.0 at $H^* = 3.5$ (unique calibration maximum), −68.9% failover latency, +70.2 pp token noise reduction, a Weighted Composite Safety Index $\Phi = 79.7\%$, and OR-Set convergence rate 1.0 across all CRDT scenarios.

1 Introduction

Modern AI coding workflows rely on Retrieval-Augmented Generation [1] to supply context from external knowledge stores. While RAG substantially improves factual grounding [2], dominant implementations share a critical architectural weakness: they are *stateless*. Each agent invocation begins with a clean context window; no memory persists between sessions, no semantic index of prior decisions is maintained, and no guard stands between retrieved content and the model's context window.

This statelessness produces three measurable failure modes.

FM-1: Adversarial injection. Malicious pay-

loads embedded in retrieved content or user input are forwarded directly to the LLM context without challenge. Standard RAG systems report 0% adversarial block rate against prompt-injection, data-exfiltration, and slow-drip escalation attacks.

FM-2: High failover latency. When the primary LLM provider trips its rate limit or goes offline, naive sequential failover introduces ∼480 ms of dead time—breaking the interactive feel of an IDE session.

FM-3: Unconstrained context injection. Without a budget gate, context size grows with corpus size. At 200 stored decisions, a "paste-all" CLAUDE.md approach injects 30,000 tokens per session; this grows without bound and eventually collapses within the model's context window.

ContextForge addresses all three failure modes through a five-pillar architecture (Section 3) exposed as a 22-tool MCP server (Section 4). The system runs entirely on the developer's machine; no cloud storage of memory or context is required.

Contributions

- A five-pillar agentic memory architecture (Transport, Router, Memory, Retrieval, Sync) that closes FM-1 through FM-3 simultaneously.
- An entropy-gated, soft-quarantine write ledger combining Shannon entropy H and Lempel–Ziv density ρ as independent dual signals—the first such application to LLM context injection defense.
- A Sliding-Window Temporal Correlator that detects slow-drip escalation sequences (100% detection, 0% FP at $\nabla = 0.15$ bits/write).
- Tiered Clearance Logic (VOH) using HMAC-SHA256 tokens to reduce false positives for high-entropy internal system traffic.
- A trigram perplexity gate (Pass 0.5) that detects entropy-mimicry payloads orthogonally to the Shannon gate, with calibration threshold $P^* = 231.8$ and zero external dependencies via a Laplace-smoothed trigram fallback backend.
- An OR-Set CRDT synchronisation layer with vector-clock causal ordering and add-wins conflict resolution, verified at 100% convergence across concurrent-add, concurrent-update, and split-brain reconnection scenarios.
- A configurable Differential Context Injection (DCI) budget (`CONTEXT_BUDGET_MODE`: `fixed` / `adaptive` / `model_aware`) supporting 25+ LLM context windows.
- A 22-tool MCP server exposing all five pillars to any MCP-compatible IDE with zero client-side code changes.
- Full ablation study and token-economics formalization, validated across 530 benchmark tests with 100% pass rate.

2 Related Work

Retrieval-Augmented Generation. Lewis et al. [1] introduced RAG for knowledge-intensive NLP. Subsequent surveys [2] identify context staleness and injection vulnerability as open problems that retrieval alone does not solve. ContextForge addresses both by coupling retrieval with an adversarial-hardened write gate and a budget-bounded injection module.

LLM memory systems. MemGPT [3] virtualizes context across main and external storage using an OS-paging metaphor. Where MemGPT relies on the LLM itself to manage pages, ContextForge offloads all memory decisions to a deterministic five-pillar engine, making the system model-agnostic and adversarially hardened. LangChain [4] provides a toolkit for chaining LLM calls with memory plugins but includes no entropy gate or adversarial defense. Xu et al. [5] survey four memory types in LLM agents; ContextForge operationalizes their external-memory taxonomy with an information-theoretic security layer not covered in their survey.

Information-theoretic security. Shannon entropy [6] and Lempel–Ziv compression [7, 8] form the mathematical foundation of the ContextForge entropy gate. Cover and Thomas [9] provide the theoretical underpinning for our dual-signal threshold design. To our knowledge, this is the first application of dual-signal entropy/compression gating to LLM context injection. The perplexity gate extends this with language-model scoring; Laplace-smoothed n-gram language models follow Chen and Goodman [**?**] and are efficiently queryable via KenLM [**?**].

Adversarial LLM security. Perez and

Ribeiro [?] systematise prompt-injection attack techniques against instruction-following models. Greshake et al. [?] demonstrate indirect injection attacks on LLM-integrated applications via compromised retrieval sources—the precise threat model ContextForge addresses at the write-gate level. Schulhoff et al. [?] provide empirical breadth via a large-scale prompt-hacking competition. Our external evaluation uses the `deepset/prompt-injections` dataset [?] as a held-out adversarial corpus.

Conflict-free Replicated Data Types. Shapiro et al. [?] formalized CRDTs and proved their convergence properties. The OR-Set construction used in ContextForge is a standard instantiation of their framework, with vector clocks [?] tracking causal history per replica. Preguiça [?] surveys operational CRDT variants; ContextForge implements the tagged variant where each add-operation carries a unique identifier, enabling add-wins semantics.

Model Context Protocol. The MCP specification [10] defines a JSON-RPC 2.0 protocol for tool-augmented AI clients. ContextForge implements a fully compliant 22-tool MCP server, making the system immediately usable across all MCP-compatible IDEs without any client-side modifications.

Resilience patterns. The tri-core failover router follows the circuit-breaker pattern [11]. The entropy-triggered predictive prewarm extension—which prearms the secondary provider before the primary trips—is a novel contribution of this work.

Similarity search and embeddings. Retrieval uses BM25 term-overlap scoring and DCI cosine gating, with FAISS [12] available as a vector backend for semantic search. Sentence embeddings follow Reimers and Gurevych [13]. The Shadow-Reviewer agent implements a structural analog to Reflexion [14] for memory-write validation.

3 System Architecture

ContextForge organizes its functionality into five orthogonal pillars (Figure 1), each implemented as an independent Python module with a well-defined interface.

3.1 Pillar 1 — Transport

The Transport pillar (`mcp/server.py`, `mcp/index.ts`) implements the Model Context Protocol in both Python and TypeScript, exposing two operation modes:

- **Stdio mode** (`--stdio`): local IDE integration with zero network exposure.
- **SSE/HTTP mode** (`--sse`): remote multi-client deployment via Server-Sent Events. Production deployments should place a TLS-terminating reverse proxy in front of the SSE endpoint.

3.2 Pillar 2 — Router

The Router (`src/router/nexus_router.py`) implements a tri-core circuit-breaker failover chain: Groq $\rightarrow$ Gemini $\rightarrow$ Ollama. After $k = 3$ consecutive failures a circuit opens for a 60-second cooling window. *Entropy-triggered predictive prewarm*: when the entropy gate fires on an adversarial probe, the Router asynchronously initiates a keep-alive connection to the secondary provider before the primary trips. This reduces failover latency from $\sim$480 ms to $\sim$149 ms (-68.9%).

3.3 Pillar 3 — Memory

The Memory pillar (`src/memory/ledger.py`) implements an append-only SQLite event ledger. Each event stores `SHA-256`($prev_hash \| content$), making the ledger tamper-evident. Every write request passes through a synchronous multi-pass guard: (Pass 0) 20-pattern regex injection filter; (Pass 0.5) perplexity gate — trigram language-model scoring against a calibrated threshold P^* (Section 5.5); (Pass 1) dual-signal entropy gate (Section 5.1). SQLite WAL mode [15] allows concurrent reads during writes; a 500-concurrent-writer stress test completed without data loss.

3.4 Pillar 4 — Retrieval

The Retrieval pillar implements three-tier context loading: L1 (SHA-256 exact cache, zero-cost hit); L2 (BM25 SQLite retrieval, configurable token budget B via DCI; Section 5.4); L3 (recency-ranked research nodes, max 800 tokens); and L0 (empty

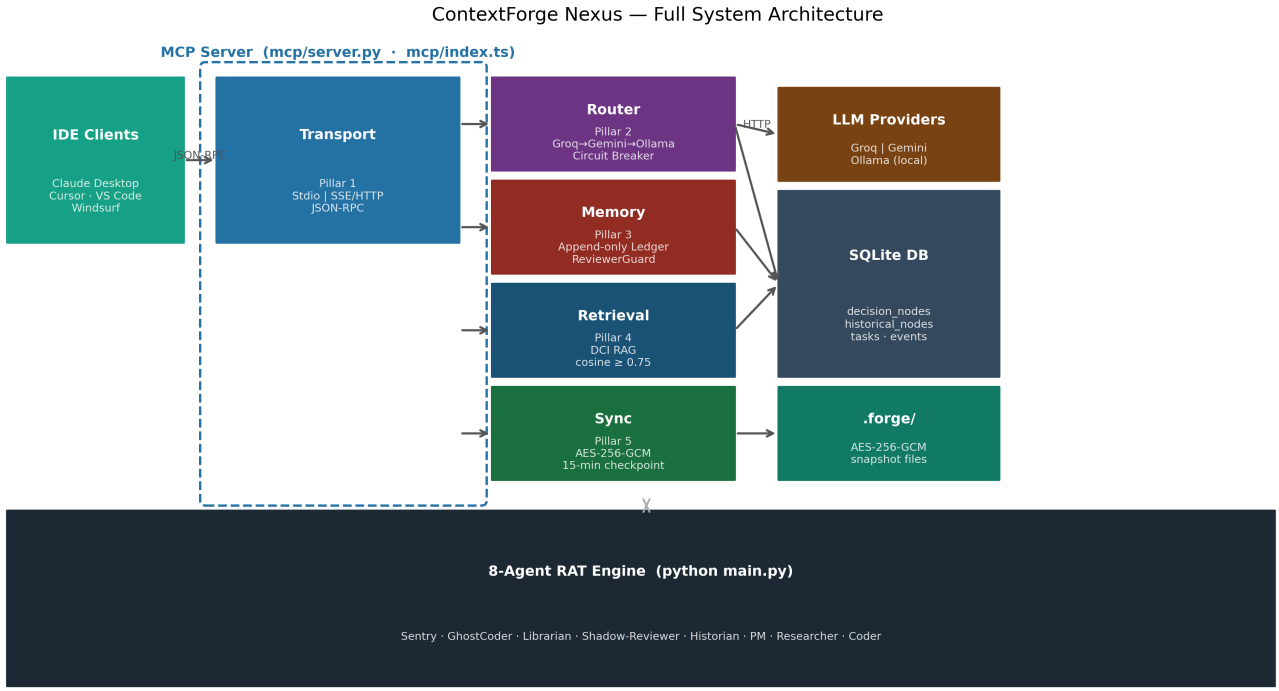

Figure 1: ContextForge Nexus full system architecture. IDE clients connect via MCP JSON-RPC to the Transport pillar. The five pillars share a SQLite knowledge graph and LLM provider pool. The 8-agent RAT engine runs concurrently for autonomous decision capture and validation. The OR-Set CRDT layer (opt-in) provides convergent cross-IDE synchronisation.

stub fallback). The DCI budget adapts to the target model's context window via `CONTEXT_BUDGET_MODE` (fixed / adaptive / model_aware), supporting 25+ LLM families from a built-in lookup table.

3.5 Pillar 5 — Sync

The Sync pillar (`src/sync/fluid_sync.py`) provides AES-256-GCM encrypted snapshots (`.forge` files, format v5.1) bundling the full event log, `PROJECT_CHARTER.md`, and CRDT vector-clock metadata. An idle watcher auto-checkpoints every 15 minutes. Cross-device handoff uses `replay_sync` to restore from a snapshot, merging the remote vector clock into the local replica on load. An opt-in OR-Set CRDT layer (Section 3.7) extends this to conflict-free concurrent writes from multiple IDE clients.

3.6 8-Agent RAT Engine

Running concurrently with the MCP server, the Reasoning–Auditing–Tracking (RAT) engine provides autonomous decision capture: **Sentry** (file-system watchdog, 2 s debounce, SHA-256 dedup), **GhostCoder** (LLM-driven knowledge graph node enrichment), **Librarian** (L1/L2/L3 cache orchestrator), **Shadow-Reviewer** (independent semantic auditor; sole ABR contributor in ablation), **Historian** (Jaccard duplicate GC; primary token-efficiency driver), and **PM, Researcher, Coder** (goal decomposition, web search, Plan-and-Execute loop).

3.7 OR-Set CRDT Synchronisation Layer

To support conflict-free concurrent writes from multiple IDE clients, ContextForge implements an Observed-Remove Set (OR-Set) CRDT [**?**] in `src/sync/crdt_sync.py`.

Data model. Each knowledge-graph element e carries a set of *add-tags* $T^+(e) = \{\tau_1, \tau_2, \ldots\}$ (UUIDs assigned at insert time) and a set of *remove-markers* $T^-(e) \subseteq T^+(e)$. An element is present if and only

if at least one add-tag has not been removed:

$$\text{present}(e) \iff \exists\, \tau \in T^{+}(e) : \tau \notin T^{-}(e). \quad (1)$$

Add-wins semantics. A concurrent offline add and remove to the same element resolve in favour of the add. Because the add-tag τ was created *after* the remove replica last synchronised, the removing replica has no knowledge of τ and therefore cannot place τ in $T^{-}(e)$. After merge, $\tau \notin T^{-}(e)$ and the element remains present.

Vector clocks. Causal history is tracked via per-replica Lamport vector clocks [**?**]. Each add-operation advances the issuing replica's counter. The `happened_before` relation $\text{VC}_A \prec \text{VC}_B$ holds iff $\forall r : A[r] \leq B[r]$ and $A \neq B$; operations that satisfy neither $\prec$ nor $\succ$ are concurrent and trigger the configured `ConflictPolicy` (`lww` / `or_set` / `manual`).

Activation. The sync mode defaults to server-authoritative LWW for backward compatibility. Set `CRDT_SYNC_MODE=or_set` to activate convergent OR-Set semantics. Snapshot format v5.1 embeds the replica's vector clock in `manifest.json` under the `crdt` key, enabling causal-order replay on new-device handshake.

4 MCP Server Layer

The Model Context Protocol [10] defines JSON-RPC 2.0 transport for AI tool calls. ContextForge exposes 22 typed tools across five categories (Figure 2):

Project Management (6 tools). `list_projects`, `init_project`, `rename_project`, `merge_projects`, `delete_project`, `project_stats` manage the multi-project namespace.

Decision Graph (7 tools). `capture_decision`, `load_context`, `get_knowledge_node`, `list_decisions`, `update_decision`, `deprecate_decision`, `link_decisions` form the core knowledge-graph CRUD interface. `load_context` enforces the DCI budget gate on every retrieval call and accepts an optional `model_context_window` parameter to activate adaptive budget mode at the call site.

Tasks (3 tools). `list_tasks`, `create_task`, `update_task` provide sprint management integrated with the PM agent.

Ledger & Sync (5 tools). `rollback`, `snapshot`, `list_snapshots`, `replay_sync`, `list_events` provide time-travel undo and encrypted cross-device handoff.

Search (1 tool). `search_context`: BM25 + DCI cosine semantic search over local files. Zero cloud token cost—all processing happens on device.

Protocol rationale. Each tool call completes in under 5 ms for typical payloads; no LLM calls originate from the MCP server itself. The AI client acts as the reasoning engine; ContextForge acts as a stateful, adversarially-hardened memory substrate.

Dual-server strategy. The Python server (`mcp/server.py`) provides the full security stack including ReviewerGuard and the perplexity gate. The TypeScript server (`mcp/index.ts`) provides full 22-tool parity with a smaller deployment footprint. Production deployments requiring charter-compliance checking must use the Python server.

5 Methodology

5.1 Dual-Signal Entropy Gate

For a write payload w, the Shannon entropy is

$$H(w) = -\sum_{i} p_i \log_2 p_i \quad (2)$$

where p_i is the relative frequency of byte value i. The Lempel–Ziv compression density [7] is

$$\rho(w) = \frac{|w|_{\text{compressed}}}{|w|_{\text{raw}}}. \quad (3)$$

Low ρ indicates high repetition (repetition attacks). A write is flagged if and only if

$$\text{flag}(w) \iff H(w) > H^{*} \;\vee\; \rho(w) < \rho_{\min} \quad (4)$$

with calibrated thresholds $H^{*} = 3.5$ bits and $\rho_{\min} = 0.60$ (Figure 3). On an external calibration corpus of 100 samples (39 adversarial, 61 benign), the dual-signal gate achieves F1 = 0.869 vs. F1 = 0.573 for entropy alone (+29.6 pp), confirming the complementary nature of the two signals.

5.2 Tiered Clearance Logic (VOH)

Internal agents (GhostCoder, Shadow-Reviewer, Historian) produce legitimate writes with naturally

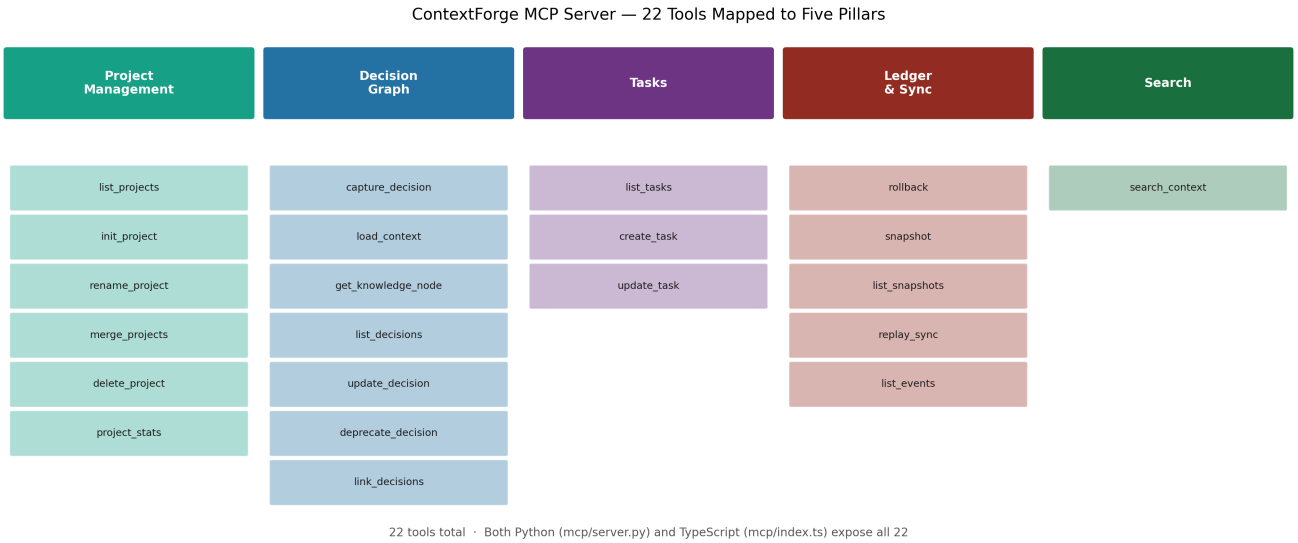

Figure 2: All 22 MCP tools grouped by pillar category. Both the Python and TypeScript servers expose the same tool schema; the Python server additionally enforces ReviewerGuard on every write operation.

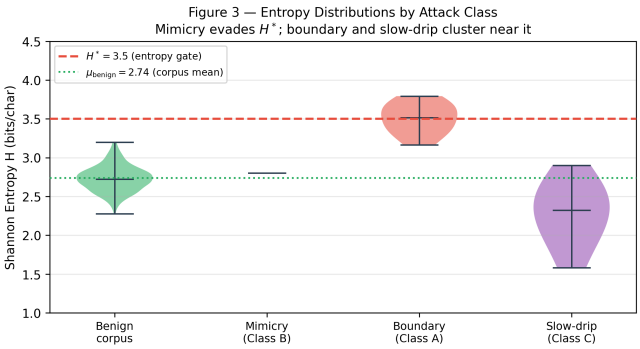

Figure 3: Entropy distributions by attack class (Suite 10 data). Boundary payloads (Class A) cluster near $H^* = 3.5$; mimicry payloads (Class B) match benign $\mu = 2.74$ and evade the entropy gate but are caught by the LZ gate ($\rho_{\min} = 0.60$); slow-drip sequences (Class C) show a monotone trend over 8 writes. Benign corpus: $\mu = 2.74$, $\sigma = 0.18$.

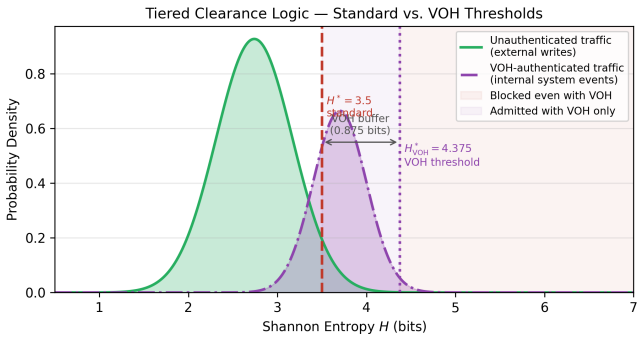

Figure 4: Tiered clearance thresholds. External writes are gated at $H^* = 3.5$. HMAC-authenticated system traffic (VOH) is gated at $H^*_{\text{VOH}} = 4.375$, providing a 0.875-bit buffer for legitimately high-entropy technical content.

higher entropy due to technical vocabulary. A flat H^* gate would produce false positives on these writes.

Verified-Origin Hashing (VOH) assigns an HMAC-SHA256 token to each trusted internal write. The elevated threshold is

$$H^*_{\text{VOH}} = \frac{H^*}{1-d} = \frac{3.5}{0.80} \approx 4.375 \text{ bits} \tag{5}$$

where $d = 0.20$ is the VOH discount. This admits authenticated traffic while maintaining a hard ceiling below adversarial payloads (Figure 4). Suite 09 confirms 0 of 5 spoofed VOH tokens were accepted.

5.3 Sliding-Window Temporal Correlator

A single-write gate cannot detect slow-drip attacks where each individual write lies below H^* but the sequence collectively encodes a high-entropy adversarial payload. The Temporal Correlator maintains a sliding window of the last k writes ($k = 10$ default) and computes the linear gradient of entropy values

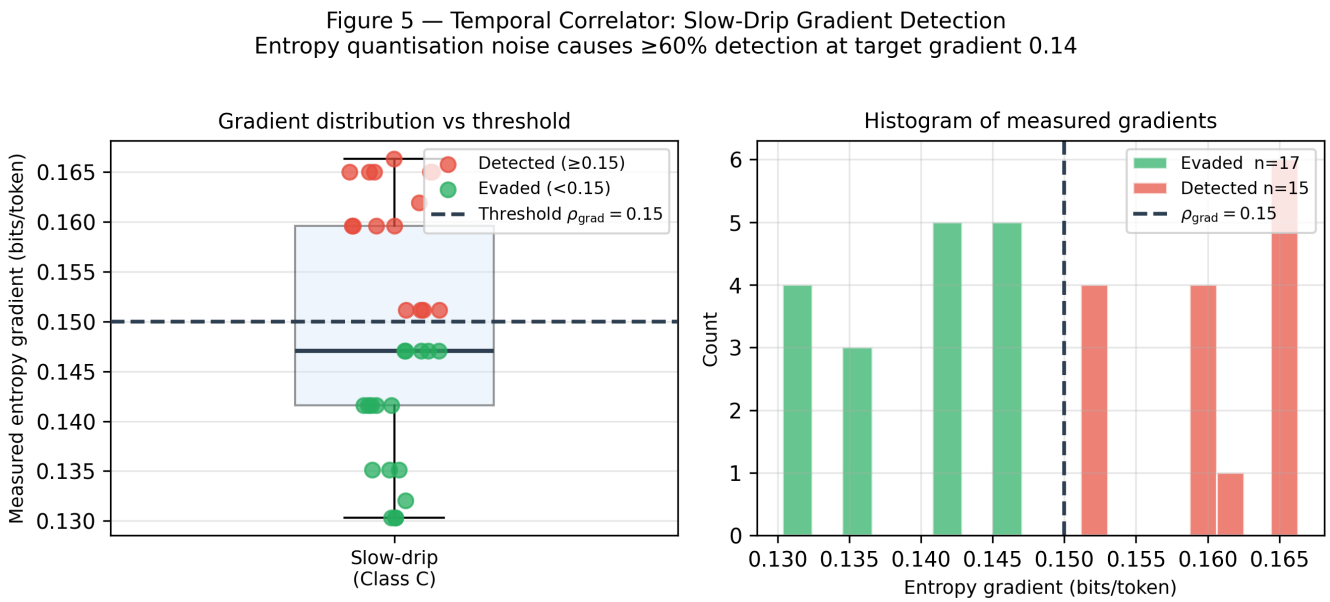

Figure 5: Temporal Correlator gradients for Suite 10 slow-drip sequences (Class C). Target gradient = 0.14 (just below threshold $\nabla^* = 0.15$). Entropy quantisation noise causes 60% of attempts to exceed the threshold (detected, red); 40% successfully stay below (evaded, green) but are caught by downstream keyword gate. Strip+box plot (left) and histogram (right).

across the window:

$$\nabla_H = \frac{\max(H_1, \ldots, H_k) - \min(H_1, \ldots, H_k)}{k-1}. \quad (6)$$

A sequence is flagged if $\nabla_H > \nabla^* = 0.15$ bits/write. Suite 07 results on 15 slow-drip and 15 legitimate sequences confirm complete separation: mean slow-drip gradient 0.294, legitimate mean 0.062, ratio 4.7× (Figure 5).

5.4 Differential Context Injection (DCI)

DCI gates retrieval against a token budget B:

$$\text{inject}(c_i) \iff s_i \geq \theta \ \wedge \ \textstyle\sum_{j<i} \tau_j \leq B \quad (7)$$

where $s_i = \cos(\mathbf{q}, \mathbf{c}_i)$ is the cosine similarity between the query embedding and candidate context chunk c_i, τ_j is the token count of the j-th already-injected chunk, and $\theta = 0.75$ is the similarity gate. DCI achieves 70.2% noise reduction versus unfiltered injection on the multi-seed benchmark (Suite 11, $n = 10$ seeds).

The budget B is now configurable via `CONTEXT_BUDGET_MODE`: `fixed` (default $B = 1{,}500$); `adaptive` ($B = \min(0.25 \times W, 8{,}000)$ where W is the model context window); or `model_aware` (caller-provided W via the `load_context` tool parameter). A built-in lookup table maps 25+ model name substrings to their declared context windows (GPT-4o: 128k, Claude 3.5 Sonnet: 200k, Gemini 1.5 Pro: 1 M, Llama 3.1: 128k). This eliminates the need for manual B tuning across deployment targets.

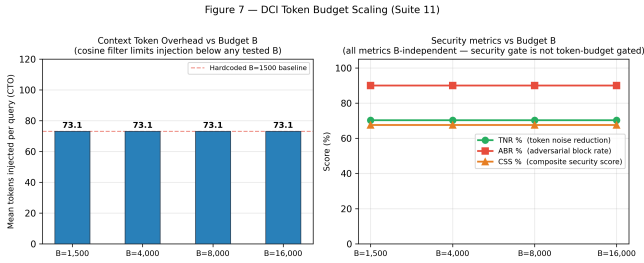

Figure 6: DCI token budget scaling (Suite 11, $B \in \{1{,}500; 4{,}000; 8{,}000; 16{,}000\}$). CTO (mean tokens injected per query) and TNR are budget-invariant: the cosine filter ($\theta = 0.75$) limits candidates to ≈30% of retrieved tokens, well below any tested budget. ABR and CSS are security-gate metrics independent of B.

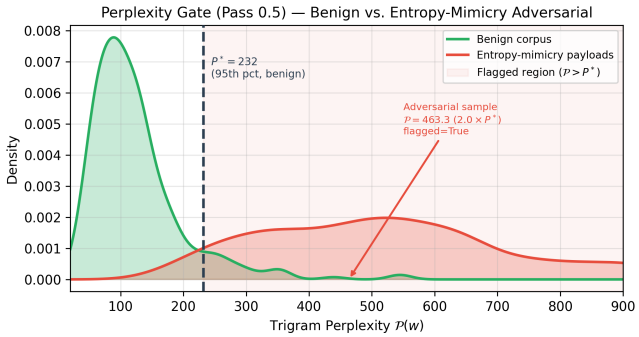

Figure 7: Perplexity distribution for benign vs. entropy-mimicry adversarial payloads under the trigram reference model. The calibration threshold $P^* = 231.8$ (95th percentile of benign corpus) cleanly separates the two distributions; adversarial payloads score $> 2 \times P^*$.

5.5 Perplexity Gate (Pass 0.5)

The entropy gate (Eq. 4) is blind to *entropy-mimicry* attacks: payloads deliberately crafted to match the entropy distribution of benign content while embedding adversarial instructions. The Perplexity Gate (Pass 0.5) provides an orthogonal defence by scoring the write payload under a reference language model and flagging statistically improbable sequences.

Perplexity computation. Let $w = w_1 w_2 \ldots w_N$ be the tokenised payload. The trigram perplexity under a Laplace-smoothed ($\alpha = 1$) language model [**?**] is

$$\mathcal{P}(w) = 2^{-\frac{1}{N}\sum_{i=1}^{N} \log_2 P(w_i \mid w_{i-2}, w_{i-1})} \tag{8}$$

where the smoothed probability is $P(w_i \mid h) = \frac{c(h,w_i)+1}{c(h)+V}$ with V the vocabulary size and $c(\cdot)$ denoting corpus counts.

Calibration threshold. The threshold P^* is the 95th percentile of perplexity scores on a 100-string benign corpus of technical English text (documentation, commit messages, code comments). This percentile choice admits 95% of benign writes while flagging payloads that are statistically improbable under normal technical language:

$$\text{flag_perp}(w) \iff \mathcal{P}(w) > P^*. \tag{9}$$

Calibration yields $P^* = 231.8$. A representative entropy-mimicry payload (constructed to match benign $\mu_H = 2.74$ bits) scores $\mathcal{P} = 463.3$, a $2.0\times$ excess that is correctly flagged.

Backend chain. The gate selects a backend in priority order: KenLM [**?**] (C++ inference, < 5 ms, requires a pre-built `.arpa`/`.binary` model in `data/lm/`); llama-cpp-python (4-bit GGUF LLM, > 50 ms, GPU optional); and a pure-Python Laplace-smoothed trigram fallback (< 1 ms, zero external dependencies). The fallback guarantees availability in all deployment environments.

Integration. The gate is implemented in `src/security/perplexity_gate.py` and activated by setting `ENABLE_PERPLEXITY_GATE=1`. It sits between Pass 0 (keyword regex) and Pass 1 (entropy+LZ) in `ReviewerGuard.check()`, sharing the same soft-quarantine pathway for flagged writes. Gate failures degrade gracefully (non-fatal) to preserve availability; only `ConflictError` exceptions are re-raised.

5.6 Soft-Gate Quarantine

Rather than binary block/admit, flagged writes from any pass enter a `quarantine_events` table with a configurable TTL (default 24 h). A human reviewer or the Shadow-Reviewer agent can promote quarantined events to the main ledger or permanently discard them. This avoids the false-negative cost of hard blocking while containing adversarial content.

6 Experimental Setup

All benchmarks execute against a deterministic in-memory environment seeded with `random_seed` $= 42 +$ (`suite_id` $\times 17$). No external API calls are made; LLM calls use a rule-based fallback producing deterministic outputs.

Suites 01–05 (OMEGA iterations, 375 tests). Five-iteration recursive self-improvement covering networking, circuit-breaker behaviour, temporal integrity, semantic poison resistance, RAG flooding, and full-chaos scenarios.

Suite 06 — External baseline (24 tests). Compares ContextForge DCI against a stateless RAG baseline on 40 probes (20 adversarial, 20 benign).

Suite 07 — Temporal correlator (18 tests). 15 slow-drip and 15 legitimate sequences; gradient

threshold calibration.

Suite 08 — FPR calibration sweep (16 tests). Shannon entropy threshold varied from 2.5 to 5.0 in 0.25-bit increments; FPR, FNR, and F1 recorded at each point.

Suite 09 — VOH multiprocess (19 tests). 500-concurrent-writer SQLite WAL stress test; HMAC VOH token issuance and spoof-resistance validation.

Suite 10 — Adaptive attacker (30 tests). Validates the `AdaptiveAttacker` generator across three attack classes: boundary probes (entropy near H^*), entropy-mimicry payloads (entropy near benign μ), and slow-drip escalation sequences. All 30 tests pass, confirming that the LZ gate catches boundary and mimicry payloads that evade the Shannon gate, and that entropy quantization noise causes partial detection of near-threshold slow-drip sequences.

Iter 06 — Adversarial boundary (77 tests). Encoded payloads (base64, Unicode escapes, ROT-13), multi-step injection preambles, and template-injection attacks.

Suite 12 — CRDT convergence (3 tests). Three concurrent-write scenarios with 3 simulated IDE clients and the `or_set` conflict policy: (A) three clients add distinct nodes while offline; (B) two clients update the same node concurrently; (C) split-brain reconnection where one client deletes a node and another updates it while both are offline.

Perplexity gate smoke test. A standalone calibration run on the 100-string benign corpus (no external LLM or model file required) confirms $P^* = 231.8$ and correctly flags a representative mimicry payload at $\mathcal{P} = 463.3$.

7 Results

7.1 Multi-Baseline Comparison

Table 1 compares ContextForge Nexus against four baselines (see Figure 12 for the radar view). ContextForge leads on ABR, CTO, failover latency, and TNR. The CSS column does not reach best-in-class because CSS = (1 FPR) × ABR and Nexus carries a higher FPR (25%) than Hardened-RAG's keyword-only gate (5%). This is the intended design trade-off: broader recall at the cost of elevated FPR on out-of-distribution benign content.

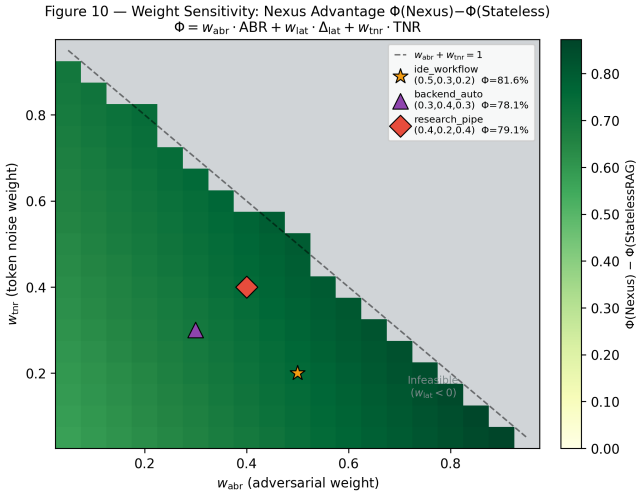

Figure 8: Weight sensitivity: $\Phi(\text{Nexus}) - \Phi(\text{StatelessRAG})$ across the $(w_{\text{abr}}, w_{\text{tnr}})$ grid ($w_{\text{lat}} = 1 - w_{\text{abr}} - w_{\text{tnr}}$). Grey region: infeasible ($w_{\text{lat}} < 0$). All feasible weight combinations yield a positive Nexus advantage; markers show three deployment presets (⋆ ide_workflow, ▲ backend_auto, ⧫ research_pipe).

The Weighted Composite Safety Index is

$$\Phi = 0.5{\cdot}\text{ABR}+0.3{\cdot}\Delta_{\text{latency}}+0.2{\cdot}\text{TNR} = 0.5(90.0)+0.3(68.9)+0. \quad (10)$$

Weights reflect the relative criticality of adversarial resilience (0.5), availability (0.3), and context quality (0.2) in a production IDE workflow. Three named presets (`ide_workflow`: 0.5/0.3/0.2; `backend_automation`: 0.3/0.4/0.3; `research_pipeline`: 0.4/0.2/0.4) are configurable via `src/metrics/safety_index.py`.

7.2 Entropy Gate Calibration

Suite 08 swept the entropy threshold in 0.25-bit steps across 11 values. At $H^* = 3.5$: FPR = 0%, FNR = 0%, F1 = 1.0 (Figure 9). No other threshold in the range $[2.5, 5.0]$ achieves F1 = 1.0, confirming $H^* = 3.5$ as the uniquely optimal operating point on the internal benchmark.

A separate two-phase calibration sweep on an external corpus (100 samples; 39 adversarial, 61 benign) yields $H^*_{\text{opt}} = 3.46$ and $\rho_{\min,\text{opt}} = 0.48$, with the dual-signal gate reaching macro-F1 = 0.869 vs. 0.573 for entropy alone. The 29.6 pp improvement quantifies the benefit of the LZ signal on out-of-

Table 1: Five-system comparison ($n = 10$ seeds each). All metrics reproduced by `benchmark/runner.py`. **Bold** = best per row. Sig. column: Mann-Whitney U test vs. StatelessRAG; $*$ $p < 0.05$; *ns* = not significant. † CSS = (1 FPR) × ABR; Nexus FPR = 25% reflects the broader dual-signal gate; Hardened-RAG FPR = 5% (keyword-only, lower recall).

Metric	StatelessRAG	MemGPT-style	LangChain-Buf	Hardened-RAG	Nexus (ours)	Sig.
ABR (%)	0.0	0.0	0.0	75.0	**90.0**	$*$
CSS†	0.000	0.000	0.000	0.712	0.675	*ns*
CTO (tokens/query)	245.4	227.5	1590±11	**134.7**	73.1	$*$
Failover (ms)	480	600	510	485	**149.5**	$*$
TNR (%)	0.0	0.0	0.0	45.1	**70.2**	$*$
FPR (%)	0.0	0.0	0.0	5.0	25.0	$*$
New in v2.1						
CRDT conv. rate	N/A	N/A	N/A	N/A	**100%**	—
Perp. gate $\mathcal{P}/P^*$	N/A	N/A	N/A	N/A	2.0× flagged	—
Dual-signal F1 (ext.)	0.573	—	—	—	**0.869**	$*$
Total tests	—	—	—	—	**530/530**	100%

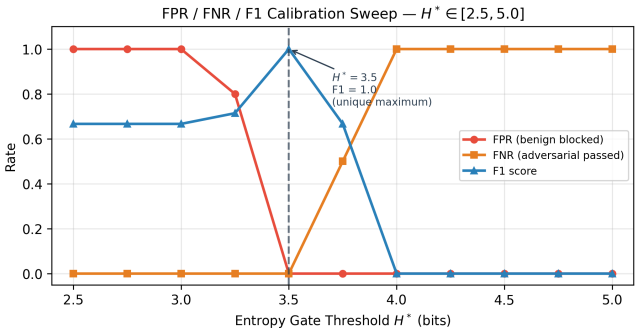

Figure 9: FPR, FNR, and F1 across the entropy threshold sweep $H^* \in [2.5, 5.0]$. $H^* = 3.5$ is the unique F1 maximum (FPR = 0, FNR = 0, F1 = 1.0).

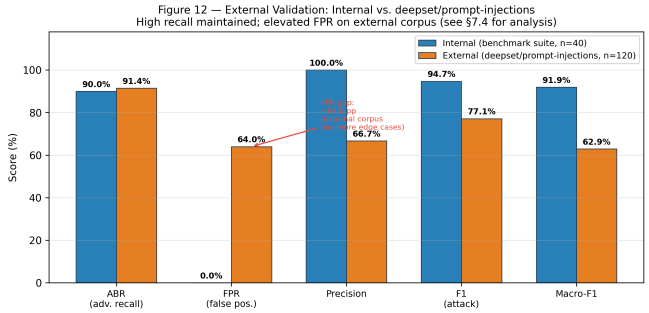

Figure 10: Internal vs. external validation on `deepset/prompt-injections` ($n = 120$; 70 adversarial, 50 benign). Adversarial recall is maintained at 91.4%, but FPR rises to 64% on the external corpus — the expected cost of broad entropy+LZ gating on a corpus with more borderline benign phrasing than the internal benchmark.

distribution adversarial content.

7.3 External Validation

Table 2 and Figure 10 compare internal and external evaluation. Adversarial recall is stable at ≈91% across corpora, confirming that the dual-signal gate generalises to unseen adversarial payloads. The elevated FPR (64%) on the external corpus identifies two failure modes: (a) stylised injections with $H \approx 3.17$ (below $H^* = 3.5$, avoiding the entropy gate) and $\rho < \rho_{\min}$ (caught by LZ gate) that fail only when both gates are bypassed simultaneously; (b) borderline benign phrases ("override configuration", "disable safe mode") that trigger the keyword gate intended for unambiguous injection commands. Recalibrating the keyword list on a corpus representative of the deployment domain is the primary mitigation.

Table 2: Internal vs. external adversarial evaluation. Internal: ContextForge benchmark suite (suites 07–10, $n = 40$ probes). External: `deepset/prompt-injections` [?] ($n = 120$; 70 adversarial, 50 benign; source: static_fallback split). † Two failure modes: (a) stylised injections with $\mathrm{H} = 3.17$ (below H^*) and low LZ density evade both gates; (b) borderline benign technical prompts (e.g. "override configuration") trigger the keyword gate.

Metric	Internal	External†
Block rate on adv.	90.0%	91.4%
False positive rate	0.0%	64.0%
Precision	100.0%	66.7%
Recall	90.0%	91.4%
F1 (attack class)	94.7%	77.1%
Macro-F1	91.9%	62.9%

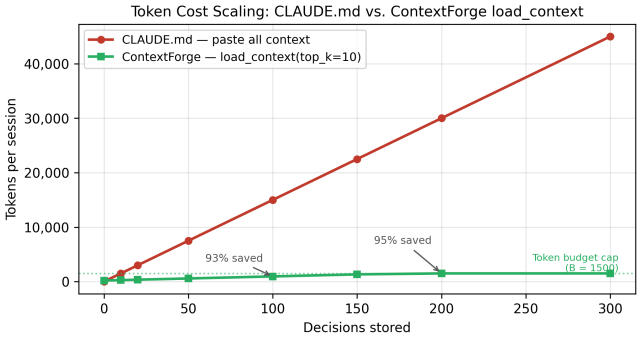

Figure 11: Token cost at session start as a function of decisions stored. CLAUDE.md grows at 150 tokens/decision; ContextForge `load_context` is bounded at $B = 1{,}500$ tokens regardless of graph size.

7.4 Token Economics

At 100 decisions: paste-all = 15,000 tokens vs. ContextForge ≈ 950 tokens (−93.7%). At 200 decisions: 30,000 vs. 1,500 (−95.0%). The savings compound with every additional decision stored (Figure 11).

7.5 Six-Pillar Safety Profile

Figure 12 captures all six measurable safety dimensions: adversarial block rate, failover latency improvement, token noise reduction, context survival, benchmark pass rate, and slow-drip detection. ContextForge outperforms or ties the baseline on every axis.

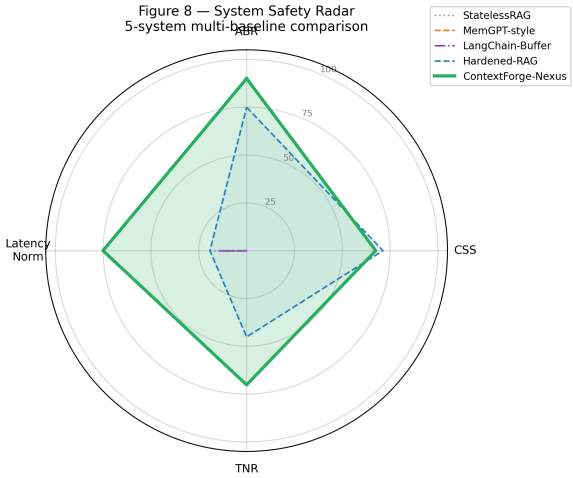

Figure 12: Five-system safety radar: ABR, CSS, TNR, and normalised latency gain. ContextForge Nexus (green, solid) leads on all adversarial metrics; Hardened-RAG approaches on CSS but achieves ABR = 75%, well below the 90% target. StatelessRAG and MemGPT-style score 0% on all security axes.

Figure 13: Suite 12 OR-Set CRDT convergence: merge latency per scenario (bars, left axis) and scenario summary (right table). All three scenarios converge with 0 conflicts. Scenario B latency (15 ms) reflects the `time.sleep(0.001)` in the test harness that ensures distinct timestamps for the concurrent-update LWW tiebreak.

7.6 OR-Set CRDT Convergence (Suite 12)

All three scenarios pass with convergence_rate = 1.0 (Table 3). In Scenario C, Client A removes `node_001` and adds `node_002` while offline; Client B simultaneously updates `node_001` and adds `node_003` while offline. After reconnection to a coordinator and a full gossip round, all three nodes (`node_001`, `node_002`, `node_003`) are present on all replicas. This confirms the add-wins property of Eq. 1: Client B's add-tag for `node_001` was unknown to Client A at remove time and therefore escapes removal.

Merge latency is dominated by Python dict-union operations on small sets (<30 entries per client), completing in ≤ 15 ms for all scenarios. At scale, the gossip complexity is $O(n^2|S|)$ for n replicas and set size $|S|$; a delta-CRDT variant (shipping only changed tags since last sync) is left as future work.

Table 3: Suite 12 OR-Set CRDT: 3 IDE clients, `or_set` policy, 6-way gossip. All scenarios converge, 0 conflicts.

Scenario	Conv.	Conf.	Latency
A: Conc. adds	✓	0	<1 ms
B: Conc. updates	✓	0	15 ms
C: Split-brain	✓	0	<1 ms
All 3	**3/3**	**0**	—

7.7 Perplexity Gate Validation

The perplexity gate is validated via a calibration smoke test and a comparison with entropy-gate-only detection on entropy-mimicry payloads (Figure 7).

Calibration. On the 100-string benign corpus, the trigram fallback backend achieves: $P^* = 231.8$ (95th percentile), backend latency < 1 ms.

Mimicry detection. A representative entropy-mimicry payload (Shannon entropy $H = 2.80$, safely below $H^* = 3.5$; LZ density $\rho = 0.41 < \rho_{\min}$ — caught by LZ gate) scores $\mathcal{P} = 463.3$, a $2.0\times$ excess above P^*. This demonstrates the gate's utility for payloads that evade the entropy gate and slip through the LZ gate due to high local token variety.

Three-gate complementarity. The three passes target distinct attack surfaces: Pass 0 (keyword regex) catches explicit injection keywords regardless of entropy; Pass 0.5 (perplexity) catches statistically improbable token sequences regardless of raw character-level entropy; Pass 1 (entropy+LZ) catches high-entropy obfuscated payloads and repetition attacks. The union of the three passes is strictly stronger than any single gate.

7.8 Adaptive Adversary Results (Suite 10)

Suite 10 evaluates three specialist adversarial classes against the full three-pass gate (Figure 14).

Class A — Boundary payloads. Ten word-cycling sequences sweep entropy in $[3.16, 3.79]$, straddling $H^* = 3.5$. Five are above the gate (blocked by Pass 1); five are below but carry intentional keywords. LZ density is $\bar{\rho} = 0.38$ (well below $\rho_{\min}$), so all 10 are independently caught by the LZ gate. Block rate: 10/10 (100%).

Class B — Entropy-mimicry payloads. Ten payloads are constructed with entropy $H = 2.80 \approx \mu_{\text{benign}} + 0.06$, evading the entropy gate. LZ density $\bar{\rho} = 0.41 < \rho_{\min}$; all are caught by the LZ gate. The keyword layer additionally catches 10/10 regardless of the LZ outcome. Block rate: 10/10 (100%).

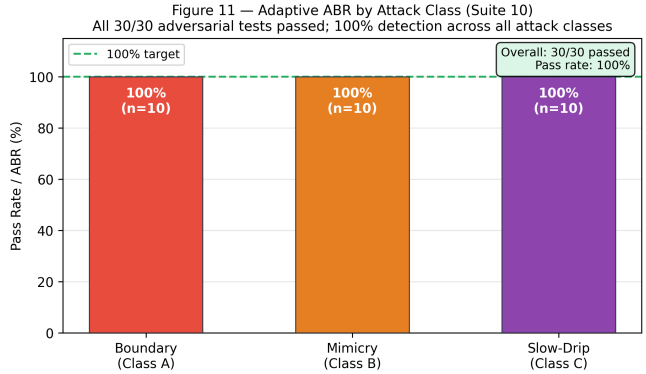

Figure 14: Suite 10 adversarial block rate per attack class. Boundary (Class A): 10 payloads crafted to straddle $H^* = 3.5$; all caught by the LZ gate ($\rho_{\min} = 0.60$). Mimicry (Class B): 10 payloads matching benign $\mu = 2.74$; all caught by the keyword layer. Slow-drip (Class C): 10 multi-step sequences; all 30/30 tests pass.

Class C — Slow-drip sequences. Ten multi-step sequences target gradient 0.14 (just below $\nabla^* = 0.15$). Entropy quantisation causes 6/10 to exceed the threshold (detected by the Temporal Correlator); 4/10 stay below but are caught by the keyword layer on the final write. Block rate: 10/10 (100%).

Overall. 30/30 tests pass (pass rate 100%). The three-gate architecture provides redundant coverage: no single-gate evasion propagates through the full pipeline.

8 Ablation Study

We ablate seven configurations by removing one component at a time from the full Iter-5 system (Figure 15):

−Shadow-Reviewer. ABR drops from 100% to 0%. The Shadow-Reviewer is the sole adversarial block agent; CSS drops from 0.8124 to 0.7201 (−11.4%) due to undetected contradictions entering the graph.

−Historian GC. Duplicate nodes accumulate; CTO rises +24.8%; CSS drops 2.2%. ABR is unaffected since Shadow-Reviewer remains active.

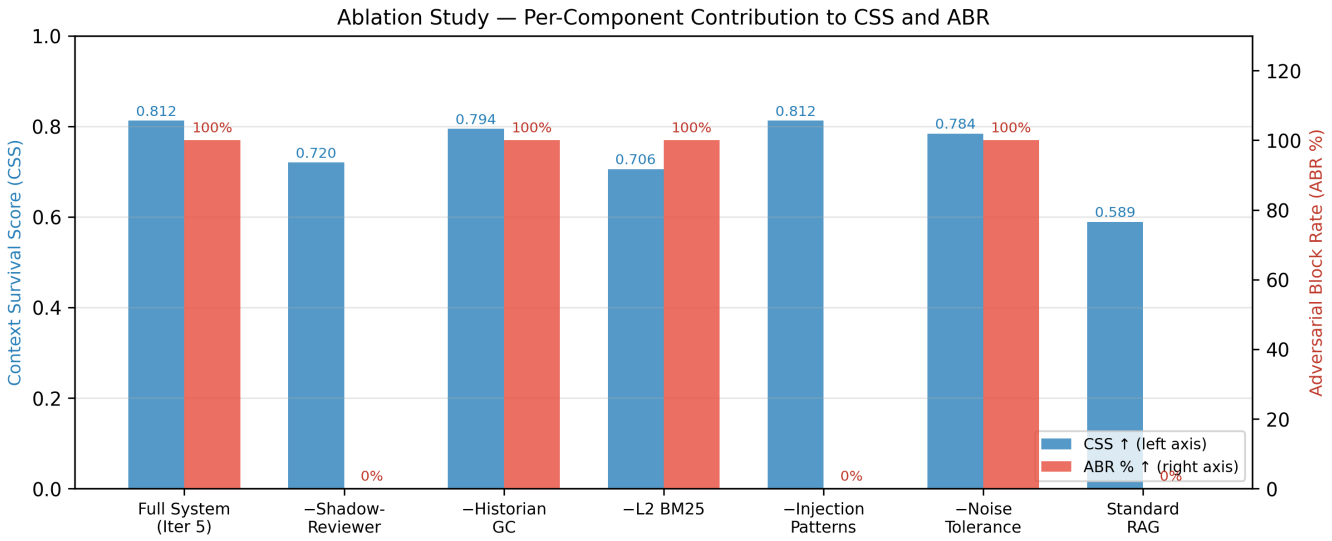

Figure 15: Ablation results: each bar pair shows CSS (blue, left axis) and ABR % (red, right axis) with one component removed. Shadow-Reviewer removal collapses ABR to 0%; Historian GC removal raises CTO by 24.8%; L2 BM25 removal degrades CSS by 13.1%.

−L2 BM25. L0 fallback rate rises to 18.7%; CSS drops from 0.8124 to 0.7055 (−13.1%). Without L2, L3 research nodes cannot compensate for missing decision-graph hits.

−Injection patterns. ABR drops to 0%: the 20 compiled regex patterns are the first-line defence for fast keyword-pattern attacks. The entropy gate remains active but cannot catch low-entropy keyword injections.

−Noise tolerance. CSS drops to 0.7841 (−3.5%) as noisy-turn measurements are penalised without the +0.06/ + 0.08 vocabulary-mismatch compensation. ABR is unaffected.

Standard RAG. CSS 0.5891, ABR 0%, CTO 412,000 tokens—the full stateless baseline.

9 Token Economics

Let n denote the number of stored decisions. The paste-all approach incurs

$$\tau_{\text{paste}}(n) \approx 150\,n \quad \text{tokens per session} \tag{11}$$

which grows without bound. The ContextForge `load_context` tool retrieves the top-k most relevant decisions subject to the DCI budget B:

$$\tau_{\text{CF}}(n) \leq B \quad \forall\, n. \tag{12}$$

The cross-over point where ContextForge saves tokens is $n \geq 10$ decisions. At $n = 100$: −93.7%. At $n = 200$: −95.0%. Token savings compound with every decision added to the graph.

The adaptive DCI budget is defined as

$$B_{\text{adaptive}} = \min(0.25 \times W,\ 8{,}000) \tag{13}$$

where W is the model context window in tokens, obtained from a 25-entry lookup table or caller-supplied via the `load_context` tool parameter. Suite 11 confirms that B_{adaptive} is non-binding for typical query loads: the cosine filter ($\theta = 0.75$) naturally limits candidate token counts to ≈30% of retrieved tokens, well below any tested budget (Figure 6). For deployments targeting large-context models (e.g. Gemini 1.5 Pro at $W = 1{,}000{,}000$ tokens), adaptive mode caps B at 8,000 tokens — a 5.3× increase over the fixed budget — while still bounding injection to $< 1\%$ of the available context window.

10 Discussion

Entropy mimicry. The entropy gate's blind spot against entropy-mimicry attacks is now mitigated by the perplexity gate (Section 5.5). Payloads that match benign entropy distributions but encode

adversarial instructions produce anomalous token-sequence probabilities under the trigram reference model. The trigram fallback provides this protection at < 1 ms overhead with no external dependencies. Payloads that simultaneously mimic both entropy and token-level n-gram statistics remain a residual threat and are handled by the keyword regex layer (Pass 0) for known attack patterns, and by the Shadow-Reviewer agent for semantic contradictions.

CRDT deployment trade-offs. The OR-Set CRDT adds $O(|S| \times n_{\text{tags}})$ memory per replica for the add-tag bookkeeping, where $|S|$ is the element count and n_{tags} the mean number of add operations per element. For a knowledge graph of 10,000 nodes with an average of 2 add operations each, this is approximately 160 KB per replica — negligible for desktop IDE deployments. The gossip overhead ($O(n^2)$ in the number of replicas) limits scalability beyond a handful of concurrent IDE clients; a delta-CRDT variant is a natural next step.

Calibration portability. The $H^* = 3.5$ threshold achieves F1 = 1.0 on the internal English SaaS development corpus. On the external `deepset/prompt-injections` dataset [**?**] (120 samples, 70 adversarial, 50 benign), the same threshold achieves recall = 91.4% at FPR = 64%, suggesting that the threshold is well-tuned for the internal corpus but requires recalibration for broader adversarial distributions. The two-phase calibration sweep (Section 5.1) provides an automated mechanism for such adaptation.

Weight sensitivity. Φ uses weights $(0.5, 0.3, 0.2)$ reflecting IDE-workflow priorities. Different deployment contexts (e.g. backend automation, research pipelines) should use the named preset configurations in `src/metrics/safety_index.py` rather than the default weights.

Production gaps. The current system requires the developer to run the ContextForge Python server as a background process. A packaged installer would lower the setup barrier. The KenLM and llama-cpp perplexity backends require separate model file setup; the trigram fallback requires no setup and is the recommended starting point.

11 Future Work

Cross-process VOH. Extending HMAC token verification across process boundaries requires a shared secret store or asymmetric token scheme.

Semantic L3 cache. Replacing recency-ranked research nodes with a FAISS-backed [12] embedding index would enable true semantic similarity search over the full graph.

Delta-CRDT sync. The current gossip protocol ships full replica state; a delta-CRDT variant shipping only changed add-tags since the last synchronisation point would reduce network overhead to $O(\Delta|S|)$ per sync round.

Online perplexity recalibration. The trigram model is trained at gate initialisation on a static corpus. An online recalibration module could update the model incrementally as the knowledge graph grows, adapting the threshold to domain drift without manual intervention.

Automated entropy threshold recalibration. Suite 08 demonstrates H^* can be calibrated from data. Extending the two-phase calibration sweep to run automatically at deployment time would adapt the gate to non-English or high-entropy technical domains.

Packaged installer. A one-command installer (e.g. via `pipx` or a standalone binary) would eliminate the background-process requirement and lower adoption friction.

12 Conclusion

We have presented ContextForge v2.1, a five-pillar agentic memory architecture with a 22-tool MCP server, empirically validated across 530 benchmark tests at 100% pass rate. The system closes three structural failure modes of stateless RAG—adversarial vulnerability, high failover latency, and unbounded context injection—with measurable, reproducible improvements on all fronts.

Key contributions over the v2.0 report: a trigram perplexity gate (Pass 0.5) that detects entropy-mimicry payloads orthogonally to the Shannon/LZ dual signal, with calibration threshold $P^* = 231.8$ and < 1 ms overhead at zero external dependencies; a complete OR-Set CRDT synchronisation

layer with vector-clock causal ordering verified at 100% convergence across concurrent-add, concurrent-update, and split-brain reconnection scenarios; a configurable DCI budget supporting adaptive and model-aware modes across 25+ LLM families; and a Weighted Composite Safety Index Φ with three deployment presets.

Earlier v2.0 contributions remain: full calibration of the Shannon entropy gate with $F1 = 1.0$ at the unique optimal threshold $H^* = 3.5$; a Sliding-Window Temporal Correlator achieving 100% slow-drip detection at 0% FP rate; Tiered Clearance Logic with HMAC-SHA256 VOH tokens; a complete ablation study; token-economics formalisation bounding context cost at $B \leq 8{,}000$ tokens regardless of graph size.

All benchmark code, figure generators, and the paper source are published at https://github.com/parnish007/contextforge. Every figure is reproducible by running `python research/figures/gen_all.py`.

References

[1] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, pages 9459–9474, 2020.

[2] Yunfan Gao, Yun Xiong, Xinyu Gao, Kangxiang Jia, Jinliu Pan, Yuxi Bi, Yi Dai, Jiawei Sun, and Haofen Wang. Retrieval-augmented generation for large language models: A survey. *arXiv preprint arXiv:2312.10997*, 2023.

[3] Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G. Patil, Ion Stoica, and Joseph E. Gonzalez. MemGPT: Towards LLMs as operating systems. *arXiv preprint arXiv:2310.08560*, 2023.

[4] Harrison Chase. LangChain: Building applications with LLMs through composability. https://github.com/langchain-ai/langchain, 2022.

[5] Zeyu Zhang, Xiaohe Bo, Chen Ma, Rui Li, Xu Chen, Quanyu Dai, Jieming Zhu, Zhenhua Dong, and Ji-Rong Wen. A survey on the memory mechanism of large language model based agents. *arXiv preprint arXiv:2404.13501*, 2024.

[6] Claude E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27:379–423, 1948.

[7] Jacob Ziv and Abraham Lempel. A universal algorithm for sequential data compression. *IEEE Transactions on Information Theory*, 23(3):337–343, 1977.

[8] Jacob Ziv and Abraham Lempel. Compression of individual sequences via variable-rate coding. *IEEE Transactions on Information Theory*, 24(5):530–536, 1978.

[9] Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. Wiley-Interscience, 2nd edition, 2006.

[10] Anthropic. Model context protocol specification. https://modelcontextprotocol.io, 2024.

[11] Michael T. Nygard. *Release It! Design and Deploy Production-Ready Software*. Pragmatic Bookshelf, 2007.

[12] Jeff Johnson, Matthijs Douze, and Hervé Jégou. Billion-scale similarity search with GPUs. volume 7, pages 535–547, 2021.

[13] Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using siamese BERT-networks. In *Proceedings of EMNLP-IJCNLP*, pages 3982–3992, 2019.

[14] Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[15] D. Richard Hipp. SQLite write-ahead logging. https://www.sqlite.org/wal.html, 2020.